MA411 Notes 2

Peter Allen

4 Random variables

Definitions

Previously (doing discrete probability), we said a random variable X is simply a function from Ω to $\mathbb{R}$. The sort of question we want to ask about discrete random variables might be: what is the probability of $X = 1$?

If we are dealing with a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, where Ω is perhaps uncountable, often the answer to any question like this will be 0. The random variables we are interested in may have probability 0 of taking any given single value. To get an interesting question, we rather ask: what is the probability of $X \in [a, b]$?

In order for that question to have an answer, we need $X^{-1}([a, b])$ to be an event: to be in $\mathcal{F}$. A function where we cannot ask this question cannot be a function we want to deal with. So we restrict attention to functions where we can ask the question.

Definition 1 (Measurable space, measurable function, random variable)**.** *If $(Y, \mathcal{F}_Y)$ and $(Z, \mathcal{F}_Z)$ are any two measurable spaces (that is, $\mathcal{F}_Y$ is a σ-algebra on Y; it's a measure space where we didn't specify the measure yet), we say a function $f : Y \to Z$ is $(\mathcal{F}_Y, \mathcal{F}_Z)$-*measurable *if $f^{-1}(F) \in \mathcal{F}_Y$ for all $F \in \mathcal{F}_Z$. If we are thinking about putting a probability measure on $(Y, \mathcal{F}_Y)$ we also say f is a* random variable*.*

If $Z = \mathbb{R}$ and $\mathcal{F}_Z = \mathcal{B}(\mathbb{R})$ is the Borel σ-algebra, we say f is a (real-valued) random variable*.*

We'll very often simply say 'random variable' when we mean real-valued. Often we in fact have a probability measure on $\mathcal{F}_Y$, and generally we'll have the Borel measure on $\mathbb{R}$: but sometimes we will want to consider different (probability) measures and still remember that we have the same random variable.

When we are 'really doing probability' we will want to (conventionally) use capital letters like X for random variables. However for now (because we are really 'still doing measure theory') I'll stick to f, g for random variables, to stress that they are functions.

It's easy to check that if the *inverse image* f^{-1} of every interval is in $\mathcal{F}_Y$, then in fact the inverse image of all Borel sets are in $\mathcal{F}_Y$ too (because $\mathcal{F}_Y$ is a σ-algebra): so we haven't been overly restrictive in the definition above. It's convenient to say this in general:

Theorem 2. *Suppose $f : Y \to Z$ is a function and $(Y, \mathcal{F}_Y)$ and $(Z, \mathcal{F}_Z)$ are measurable spaces. Let $\mathcal{G}$ be any collection of subsets of Z such that $\sigma(\mathcal{G}) = \mathcal{F}_Z$. If $f^{-1}(G) \in \mathcal{F}_Y$ for*

all $G \in \mathcal{G}$, then f is a $(\mathcal{F}_Y, \mathcal{F}_Z)$-measurable function. (If we are thinking of Y as having a probability measure: it is a random variable.)

The proof is more or less definitional.

Proof. We need to show that for all $F \in \mathcal{F}_Z$ we have $f^{-1}(F) \in \mathcal{F}_Y$. We let

$$\mathcal{F}'_Z = \{F \in \mathcal{F}_Z \,:\, f^{-1}(F) \in \mathcal{F}_Y\}\,.$$

By definition $\mathcal{G} \subseteq \mathcal{F}'_Z \subseteq \mathcal{F}_Z$.

On the other hand, $\mathcal{F}'_Z$ is a σ-algebra: this is trivial since inverse image commutes with the σ-algebra operations. It follows

$$\mathcal{F}_Z = \sigma(\mathcal{G}) \subseteq \sigma(\mathcal{F}'_Z) = \mathcal{F}'_Z$$

and so $\mathcal{F}'_Z = \mathcal{F}_Z$ as desired. □

This helps us check if a given function f is really a real-valued random variable, for example: it is enough to check all intervals $(-\infty, c]$ have $f^{-1}((-\infty, c])$ an event. (Or we can use any other generating set of the Borel σ-algebra, as convenient.)

Definition 3 (σ-algebra generated by a random variable)**.** *Let $f : Y \to Z$ be a measurable function from $(Y, \mathcal{F}_Y)$ to $(Z, \mathcal{F}_Z)$. Then $\sigma(f) = \{f^{-1}(F) : F \in \mathcal{F}_Z\}$ is the* σ-algebra generated by f.

For the same reason as before, $\sigma(f)$ *is* a σ-algebra. Recalling that σ-algebras carry information, $\sigma(f)$ tells us 'the information carried by the random variable f'.

Returning to Example 10 from Section 1: suppose Y is the sample space coming from flipping a coin 10 times, and $\mathcal{F}_Y$ is the σ-algebra we describe there, which contains exactly the information of the first five flips (when we 'pause'), what does it mean to say f is a $\mathcal{F}_Y$-measurable random variable? It means that knowing the first five outcomes is enough for us to calculate f. We'll see (Theorem 12 below) that this holds in some generality.

Sometimes we may want to think about the σ-algebra generated by a collection of random variables $(f_i)_{i\in I}$, which we define to be $\sigma\big((f_i)_{i\in I}\big) := \sigma\big(\bigcup_{i\in I} \sigma(f_i)\big)$. Just as $\sigma(f)$ is the smallest σ-algebra with respect to which f is measurable, so $\sigma\big((f_i)_{i\in I}\big)$ is the smallest σ-algebra with respect to which each f_i is measurable.

Definition 4 (Sure, almost sure, almost sure equality)**.** *We often 'do not care' what happens on sets of probability zero. We say an event F in $(\Omega, \mathcal{F}, \mathbb{P})$ is* sure *if it is the event Ω, and* almost sure *if $\mathbb{P}(F) = 1$. We are usually more interested in the latter.*

Suppose f and g are two random variables on the same probability space $(\Omega, \mathcal{F}, \mathbb{P})$. We say f and g are almost surely equal *if $\{\omega : f(\omega) \neq g(\omega)\}$ has probability 0 (it is a $\mathbb{P}$-null set).*

In a general measure space $(X, \mathcal{F}, \mu)$ we would say a property holds 'almost everywhere' instead of 'almost surely': that is, if the set of $x \in X$ such that the property does not hold is a μ-null set.

In some texts 'random variable' is defined differently: a 'random variable' is an equivalence class of measurable functions, with two functions defined to be equivalent if they are almost surely equal. This is pretty in the sense that two random variables are considered different if and only if the difference 'matters': but it also means an extra level of abstraction and in some situations this abstraction is actively unhelpful.

In particular, whether two random variables are almost surely equal depends on $\mathbb{P}$. If we switch to a different probability measure $\mathbb{Q}$ on the same σ-algebra, we can have different $\mathbb{P}$-null and $\mathbb{Q}$-null sets, and f and g may be $\mathbb{P}$-almost surely equal but not $\mathbb{Q}$-almost surely equal.

The next theorem lets us do algebra and analysis with real-valued random variables. It's sometimes convenient to work with the *extended real line* $\mathbb{R}^{\pm\infty} = \mathbb{R} \cup \{-\infty, +\infty\}$ where we think of $-\infty$ as 'a point smaller than all real numbers' and $+\infty$ as 'a point larger than all real numbers'. We can extend the definition of 'limit of a sequence', and sup and inf, to this context: any non-empty set of extended real numbers has both a sup and an inf in $\mathbb{R}^{\pm\infty}$. There is a way of doing this formally by defining a topology on $\mathbb{R}^{\pm\infty}$: the usual open intervals remain open, and in addition we say $[-\infty, x)$ and $(x, +\infty]$ are open sets for all $x \in \mathbb{R}$. The topology generated by this captures the notion of limit we are interested in; its Borel σ-algebra is the 'right' extension of $\mathcal{B}(\mathbb{R})$ to the extended real line.

Theorem 5. *Let $f, f_1, \ldots$ and g be measurable functions from $(Y, \mathcal{F})$ to $(\mathbb{R}^{\pm\infty}, \mathcal{B}(\mathbb{R}^{\pm\infty}))$; let $\lambda \in \mathbb{R}$. Then, provided they are defined, the following are all measurable functions:*

$$f + g\,,\ fg\,,\ \tfrac{1}{f}\,,\ \lambda f\,,\ \inf_{n\in\mathbb{N}} f_n\,,\ \sup_{n\in\mathbb{N}} f_n\,,\ \liminf_{n\geq 1} f_n\,,\ \limsup_{n\geq 1} f_n \quad \textit{and} \quad \lim_{n\to\infty} f_n\,.$$

It's convenient to note that (as special cases of sup and inf) the functions $\max(f, g)$, $\min(f, g)$, $|f|$, $f^+ := \max(f, 0)$ and $f^- := \max(-f, 0)$, where 0 is the constant 0 function, are measurable whenever f and g are.

The 'if they are defined' applies to the algebraic operations (we cannot add ∞ and $-\infty$, or multiply 0 and ∞, or divide by 0) and to the limit.

Proof. For the 'algebraic' properties, we can use a bit of analysis knowledge. A continuous function from $\mathbb{R}^c \to \mathbb{R}^d$ is by definition Borel-measurable (the inverse image of every open set is open, so measurable; and the open sets generate the Borel σ-algebra on $\mathbb{R}^d$) and this remains true when we extend to $\mathbb{R}^{\pm\infty}$. The functions $(x, y) \to x + y$, $(x, y) \to xy$, $x \to \frac{1}{x}$ and $x \to \lambda x$ are all continuous functions. And if f and g are measurable functions from $(Y, \mathcal{F})$ to $(\mathbb{R}^{\pm\infty}, \mathcal{B}(\mathbb{R}^{\pm\infty}))$, then so is (f, g) mapping to $(\mathbb{R}^{\pm\infty})^2$. To see the last, it's enough to show that the inverse image of any $(a, b) \times (c, d)$ is $\mathcal{F}$-measurable, because (we proved in Chapter 3) these generate the Borel σ-algebra. But this inverse image is $f^{-1}((a, b)) \cap f^{-1}((c, d))$; the two sets we intersect are in $\mathcal{F}$ since f and g are measurable, so the intersection is too. Putting these facts together we obtain all four algebraic properties (and this is how to deal with algebraic properties in general).

The 'analytic' functions are the following exercise.

Exercise 1. *Prove that for any $y_1, \cdots \in \mathbb{R}^{\pm\infty}$ and any $c > -\infty$, we have $\inf_{n\in\mathbb{N}} y_n \leq c$ if and only if*

$$\forall n \in \mathbb{N} : \exists m \in \mathbb{N} : y_m \leq c + \tfrac{1}{n}\,.$$

Deduce that for any $c > -\infty$ we have

$$\{\omega : \inf_{n\geq 1} f_n(\omega) \leq c\} = \bigcap_{n\in\mathbb{N}} \bigcup_{m\in\mathbb{N}} \{\omega : f_m(\omega) \leq c + \tfrac{1}{n}\}.$$

Hence show that if $f_1, \cdots : \Omega \to \mathbb{R}^{\pm\infty}$ are measurable functions, so is $\inf_{n\in\mathbb{N}} f_n$.

Deduce that $\sup_{n\in\mathbb{N}} f_n$ is also measurable. Prove that all three remaining 'analytic' functions are measurable (assuming the pointwise limit exists). Hint: use the fact that $\liminf_{n\in\mathbb{N}} f_n = \sup_{n\in\mathbb{N}} \inf_{k\geq n} f_k$ because the sequence of infima is monotone increasing.

Solution to Exercise 1:

By definition, $\inf_{n\in\mathbb{N}} y_n \leq c$ if and only if there is no $\varepsilon > 0$ such that $c + \varepsilon$ is a lower bound for $\{y_1, \dots\}$. Equivalently, this is if and only if for all $n \in \mathbb{N}$, there is $m \in \mathbb{N}$ such that $y_m < c + \frac{1}{n}$. This immediately gives the forward implication; for the reverse implication, we replace n with $2n$ and we are told there is $y_m \leq c + \frac{1}{2n} < c + \frac{1}{n}$ as required. Observe this argument works if c is any real number, and (trivially—both statements are True) if $c = \infty$.

If $c = -\infty$, we need a slightly different statement: $\inf_{n\in\mathbb{N}} y_n = -\infty$ if and only if for all $n \in \mathbb{N}$ there exists $m \in \mathbb{N}$ such that $y_m < -n$. This is simply the definition of inf for the extended real line.

For the set equality, we simply apply the definitions of set equality, intersection and union: ω' is in the RHS set if and only if for all $n \in \mathbb{N}$, ω' is in $\bigcup_{m\in\mathbb{N}}\{\omega : f_n(\omega) \leq c + \frac{1}{n}\}$, which is if and only if for all $n \in \mathbb{N}$ there exists $m \in \mathbb{N}$ such that $\omega' \in \{\omega : f_n(\omega) \leq c + \frac{1}{n}\}$; which is precisely the quantified statement from the previous part. So the two sets are equal for any $c > -\infty$. Again, for $c = -\infty$ we get a slightly different set equality:

$$\{\omega : \inf_{n\geq 1} f_n(\omega) = -\infty\} = \bigcap_{n\in\mathbb{N}} \bigcup_{m\in\mathbb{N}} \{\omega : f_m(\omega) \leq -n\}.$$

To prove $f = \inf_{n\in\mathbb{N}} f_n$ is measurable, it is enough to show $f^{-1}(G) \in \mathcal{F}$ for all G in some generating set $\mathcal{G}$ of $\mathcal{B}(\mathbb{R}^{\pm\infty})$. The easiest generating set to take is $\mathcal{G} = \{[-\infty, c] : c \in \mathbb{R}^{\pm\infty}\}$: because we just proved that for all $c \in \mathbb{R}^{\pm\infty}$ we can write $f^{-1}([-\infty, c])$ as a countable intersection of countable unions of sets which are (by measurability of $f_1, \dots$) in $\mathcal{F}$. Since $\mathcal{F}$ is a σ-algebra, this verifies $f^{-1}([-\infty, c]) \in \mathcal{F}$ for all $c \in \mathbb{R}^{\pm\infty}$, and so by Theorem 2, f is measurable.

For the supremum, we note $\sup_{n\in\mathbb{N}} f_n = -\inf_{n\in\mathbb{N}}(-f_n)$, which is measurable. For the limit infimum, we use the hint $\liminf_{n\geq 1} f_n = \sup_{n\in\mathbb{N}} \inf_{n\geq k} f_k$, and observe that we just proved each infimum term is a measurable function, so the sup is a sup of countably many measurable functions and is therefore measurable. The limit supremum follows in much the same way: and finally if $\lim_{n\to\infty} f_n$ exists, then it is equal to the limit infimum and hence measurable.

□

For the 'analytic' properties, it's important to note that although the inf and sup are well defined functions even if one takes infimum or supremum over an uncountable set (this is obviously undefined for limit infimum et cetera!), our proof used the fact that we only took

the supremum of a countable set of functions. Indeed, the supremum of an uncountable set of functions might not be measurable: working in $\mathcal{B}(\mathbb{R})$, the function $\mathbf{1}_{\{x\}}$ (the indicator of a singleton set) is measurable for all $x \in \mathbb{R}$, but for any $S \subseteq \mathbb{R}$ we can write $\mathbf{1}_S = \sup_{x \in S} \mathbf{1}_{\{x\}}$. If S is not in $\mathcal{F}$, this won't be a measurable function.

Examples

There are a collection of 'standard' random variables which are worth knowing. We begin with some *discrete* random variables, which map from some probability space $(\Omega, \mathcal{F}, \mathbb{P})$ to a countable (or even finite) set. We stress that we don't care what the probability space is, just what the *distribution* of the random variable is (i.e. the probability of it lying in any given interval). Since we're 'doing probability' in this section, we'll use X and Y for our random variables.

We'll usually say that a random variable 'is a binomial', but sometimes 'is binomially distributed' (or the same with other distribution names).

Example 6 (Bernoulli random variable, indicator function)**.** *If $X : \Omega \to \{0, 1\}$ is any measurable function, we call it a* Bernoulli *random variable* $\mathrm{Ber}(p)$*. We often say X 'has probability p' when we mean $\mathbb{P}(X = 1) = p$. A Bernoulli random variable is an example of an* indicator function*. We often write $\mathbf{1}_A$ for the function from Ω (or any other set!) to $\{0, 1\}$ which takes value 1 exactly on A; we also write $\mathbf{1}$(the sample has Property B) (complicated set descriptions are easier not in subscripts).*

This sort of random variable might represent 'success or failure'; it's also often convenient to break one of the following more complicated 'counting' random variables into a sum of indicator random variables.

Example 7 (Binomial random variable)**.** *Given $n \in \mathbb{N}$ and $p \in [0, 1]$, if for all integer $0 \leq k \leq n$ we have $\mathbb{P}[X = k] = \binom{n}{k} p^k (1 - p)^{n-k}$, we say X is a* binomial *random variable* $\mathrm{Bin}(n, p)$*, or $X \sim \mathrm{Bin}(n, p)$ (X is distributed as $\mathrm{Bin}(n, p)$).*

This random variable counts the number of successes in n independent trials with probability p; in other words, it is a sum of n independent probability p Bernoulli random variables.

Example 8 (Poisson random variable)**.** *Given $\lambda > 0$, if for all integer $k \geq 0$ we have $\mathbb{P}(X = k) = e^{-\lambda} \cdot \frac{\lambda^k}{k!}$, we say X is a* Poisson *random variable* $\mathrm{Poi}(\lambda)$*, or $X \sim \mathrm{Poi}(\lambda)$ (X is distributed as $\mathrm{Poi}(\lambda)$).*

A Poisson random variable is the 'limit' of binomial random variables as n goes to ∞, with $np = \lambda$ kept constant; this turns out to be a natural thing to study.

Exercise 2. *Prove that for any fixed k and $\lambda > 0$, we have $\mathbb{P}(\mathrm{Bin}(n, \frac{\lambda}{n}) = k) \to \mathbb{P}(\mathrm{Poi}(\lambda) = k)$ as $n \to \infty$.*

Solution to Exercise 2:

We have by definition

$$\mathbb{P}\big(\mathrm{Bin}(n, \tfrac{\lambda}{n}) = k\big) = \binom{n}{k} \big(\tfrac{\lambda}{n}\big)^k \big(1 - \tfrac{\lambda}{n}\big)^{n-k} = \tfrac{n(n-1)\ldots(n-k+1)}{k!} \big(\tfrac{\lambda}{n}\big)^k \big(1 - \tfrac{\lambda}{n}\big)^n \big(1 - \tfrac{\lambda}{n}\big)^{-k}.$$

Recalling that k is fixed, we have $\lim_{n\to\infty} \frac{n(n-1)\dots(n-k+1)}{n^k} = 1$, we have $\lim_{n\to\infty} \left(1-\frac{\lambda}{n}\right)^{-k} = 1$, and we have $\lim_{n\to\infty} \left(1-\frac{\lambda}{n}\right)^{n} = e^{-\lambda}$, so

$$\lim_{n\to\infty} \mathbb{P}\big(\mathrm{Bin}(n, \tfrac{\lambda}{n}) = \frac{\lambda^k}{k!} e^{-\lambda} = \mathbb{P}\big(\mathrm{Poi}(\lambda) = k\big)\,.$$

We now move on to some *continuous* random variables, which we specify by giving a continuous distribution function (as in Chapter 3).

Example 9 (Gaussian random variable)**.** *Suppose for given $\mu \in \mathbb{R}$ and $\sigma > 0$ we have*

$$\mathbb{P}\big(X \in [-\infty, a)\big) = \int_{x=-\infty}^{a} \tfrac{1}{\sigma\sqrt{2\pi}} \cdot \exp\Big(-\frac{(x-\mu)^2}{2\sigma^2}\Big)\,\mathrm{d}x\,.$$

Then we say X is a Gaussian *(or* normal*) random variable $N(\mu, \sigma^2)$*

The integral is the *distribution function* of the continuous random variable (and the distribution function is continuous because we can write it as an integral!), but we often care about the integrand $\frac{1}{\sigma\sqrt{2\pi}} \cdot \exp\big(-\frac{(x-\mu)^2}{2\sigma^2}\big)$, which is the *probability density function* of the random variable.

Gaussian random variables are famously the 'limit' of shifted $\mathrm{Bin}(n,p)$ random variables, but also of many other sums of independent random variables: this is the *central limit theorem*. Warning: this 'limit' does *not* mean the same as 'limit' when we talked about Poisson random variables (and we didn't specify properly what we meant in either case).

Example 10 (Exponential random variable)**.** *Suppose for given $\lambda \in \mathbb{R}$ we have*

$$\mathbb{P}\big(X \in [-\infty, a)\big) = \begin{cases} 1 - e^{-\lambda a} & a \geq 0 \\ 0 & a < 0 \end{cases}\,.$$

Then we say X is an exponential *random variable* $\mathrm{Exp}(\lambda)$*.*

The probability density function of $\mathrm{Exp}(\lambda)$ takes value 0 on negative x and $\mu e^{-\mu x}$ for non-negative x.

Any (real valued) random variable X has a distribution function $\mathbb{P}(X \in (-\infty, a])$. For discrete random variables, this function will be a 'step function'.

5 Simple functions

As we saw in Chapter 1, something we want to do with random variables is talk about their expectation. In order to do that in general, we need a notion of *integral* which is general enough. This is the main aim of what follows in this chapter and the next (though we will use it for other things along the way). We are back to measure theory (really) at this point, so functions will be lower case latters. However, we'll stick to a probability space for our domain. All of what we will do works in a general finite measure space (i.e. $\mu(X) < \infty$), but this is, by scaling, basically the same thing as a probability space; often enough we do not want to work in a measure space where $\mu(X) = \infty$.

Definition 11 (simple function)**.** *Given a probability space* $(\Omega, \mathcal{F}, \mathbb{P})$*, for any* $A \in \mathcal{F}$ *we call the function* $\mathbf{1}_A : \Omega \to \mathbb{R}^{\pm\infty}$ simple*. We say that any multiple of a simple function, and any sum of two simple functions, is simple.*

This makes the set of simple functions a vector space. It's important to remember, though, that a simple function is always a *finite* sum of multiples of indicator functions.

While we are allowing ∞ and $-\infty$ in our codomain, this is mainly for technical convenience. The interesting cases, generally, will be when we don't use either symbol.

Exercise 3. *Suppose* f *is a simple function. Prove that we can write* f *as a finite sum of multiples of indicator functions of pairwise disjoint sets ('disjoint indicator representation'). Suppose that* f *and* g *are any two simple functions: prove that* $\max(f, g)$ *and* $\min(f, g)$ *are simple functions.*

Solution to Exercise 3:

Let $f = \sum_{i=1}^{n} \alpha_i \mathbf{1}_{A_i}$. Consider the *Venn diagram* formed by the n sets: this contains 2^n sets indexed by $I \subseteq [n]$ (recall: $[n] = \{1, 2, \ldots, n\}$), where

$$A_I = \{\omega \,:\, \omega \in A_i \iff i \in I\} = \bigcap_{i \in I} A_i \setminus \bigcup_{i \notin I} A_i$$

is a measurable set. By definition, f is constant with value $\alpha_I \geq 0$ on each A_I, so we have

$$f = \sum_{I \subseteq [n]} \alpha_I \mathbf{1}_{A_I} \,,$$

and the sets A_I are by definition pairwise disjoint.

For the maximum, we apply a similar construction: suppose f and g are simple functions, and consider disjoint indicator representations of the two functions. Without loss of generality (by adding 0 times the indicator of the remaining set) suppose in both cases the indicator functions partition Ω. For each $\alpha \mathbf{1}_A$ in the representation of f and $\beta \mathbf{1}_B$ in the representation of B, we take a term $\max(\alpha, \beta)\mathbf{1}_{A \cap B}$ (or min, respectively). The sum of these terms, which is by definition simple, is easily checked to be $\max(f, g)$ (or min, respectively): for any $\omega \in \Omega$, it is in exactly one set A in the representation of f, and exactly one set B in the representation of g, so it is in $A \cap B$ and no other pairwise intersection.

Finally (though we don't need it for this exercise!) it's convenient to record a couple more representations. Let f be a non-negative simple function; then f takes a finite set of values $0, v_1, v_1 + v_2, \ldots, v_1 + \cdots + v_t$. Let $\sum_{i=1}^{n} \alpha_i \mathbf{1}_{A_i}$ be a disjoint indicator representation of f; we can write $\alpha_i = \sum_{j=1}^{t_i} v_j$ for each i by definition, and $\sum_{i=1}^{n} \sum_{j=1}^{t_i} v_j \mathbf{1}_{A_i}$ is the *block representation* of f (we can think of this as dividing each of our disjoint indicators into 'blocks' all of the same height). We can also write $B_j = \bigcup_{i \,:\, t_i \geq j} A_i$, and we have the *slab representation* $f = \sum_{j=1}^{t} v_j \mathbf{1}_{B_j}$; here we have $B_1 \supseteq B_2 \supseteq \ldots$.

A corollary of this is that if f is a simple function which takes non-negative values, then we can write it as sum of *positive* multiples of indicator functions. Although the set of simple functions definitely does not contain all measurable functions—or even any non-constant continuous functions—any measurable function can be approximated by simple functions.

There are many ways to do this: here is one. Suppose $f : \Omega \to \mathbb{R}^{\infty}_{\geq 0}$ is any non-negative measurable function.

Let $f_0 = f$, and now for each $n \geq 1$ we do the following.

If n is odd, let $A_n = \{\omega \in \Omega : f_{n-1}(\omega) \geq 1\}$, let $\alpha_n = 1$; and let $f_n = f_{n-1} - \mathbf{1}_{A_n}$.

If n is even, let A_n be the set of $\omega \in \Omega$ such that the fractional part of $f_{n-1}(\omega)$ is at least $2^{-n/2}$, let $\alpha_n = 2^{-n/2}$; and let $f_n = f_{n-1} - 2^{-n/2}\mathbf{1}_{A_n}$. If $f_{n-1}(\omega) = \infty$, we define the fractional part of $f_{n-1}(\omega)$ to be 0.

We claim this gives the required approximation of f.

Exercise 4. *Fill in the details: Prove that this constructs a pointwise monotone increasing sequence* $(g_n)_{n\in\mathbb{N}} = \left(\sum_{i=1}^{n} \alpha_i \mathbf{1}_{A_i}\right)_{n\in\mathbb{N}}$ *which converges pointwise to* f.

Prove that if f *is a bounded function, the convergence is uniform (i.e. for any given* ε *there is* $N \in \mathbb{N}$ *such that for all* x *we have* $|f(x) - g_n(x)| < \varepsilon$*).*

Prove that for all n *we have* $\sigma(g_n) \subseteq \sigma(f)$*. (Hint: use induction.)*

Solution to Exercise 4:

That $(g_n)_{n\in\mathbb{N}}$ is pointwise monotone increasing is trivial: the difference $g_n - g_{n-1}$ is $\alpha_n \mathbf{1}_{A_n}$ which is non-negative; it follows that the sequence of functions converges pointwise to a limit function (though we will prove convergence to f directly).

Given $\omega \in \Omega$, if $f(\omega) = \infty$, for every odd n we have $\omega \in A_n$ and hence for all n we have $g_n(\omega) \geq \frac{n}{2}$, so $g_n(\omega)$ tends to $f(\omega) = \infty$ as required. If on the other hand $f(\omega)$ is a real number, let $f(\omega) = t + x$ where $0 \leq x < 1$. For each odd $1 \leq n \leq 2t - 1$, we have $\omega \in A_n$, and for odd $n \geq 2t + 1$ we have $\omega \notin A_n$. Considering x to be a binary number $0.x_1x_2\ldots$ where each x_i is equal to either 0 or 1, suppose that either $x = 0$ and $x_1 = \cdots = 0$, or $x > 1$ and there is no t such that $x_i = 0$ for all $i > t$. (A number of this form can equally well be represented, for the minimal such t, by setting $x_t = 0$ and $x_i = 1$ for all $i > t$). Then we have $\omega \in A_{2n}$ if and only if $x_n = 1$ by definition. Then the approximating sequence $g_n(\omega)$ tends to $t + x$ by construction (and since every number x has a binary representation).

If there is some absolute T such that $0 \leq f(\omega) \leq T$ for all $\omega \in \Omega$, then from above we have $A_n = \emptyset$ for all odd $n \geq 2T + 1$, and by the usual properties of a binary representation, if $n \geq 2T$ is even then $f(\omega) - g_n(\omega) \leq 2^{-n/2}$. In particular, this verifies uniform convergence for bounded functions: given $\varepsilon > 0$ we choose an even $N \geq 2T$ such that $2^{-N/2} < \varepsilon$.

Finally, we prove the claimed statement about σ-algebras by induction. We have $\sigma(g_1) = \{\emptyset, A_1, {A_1}^c, \Omega\}$ is contained in $\sigma(f)$ because $A_1 = f^{-1}\big([1, \infty]\big)$. For each $n \geq 2$, assuming $\sigma(g_{n-1}) \subseteq \sigma(f)$, to check $\sigma(g_n) \subseteq \sigma(f)$ it suffices to verify $A_n \in \sigma(f)$ (because A_n together with $\sigma(g_{n-1})$ generate $\sigma(g_n)$). Furthermore, we have f_{n-1} is $\sigma(f)$-measurable by Theorem 5 (applied with $\mathcal{F} = \sigma(f)$). By construction A_n is the inverse image of a measurable set under f_{n-1}: if n is odd, the set is $[1, \infty]$, while if n is even, the set is $\bigcup_{t\in\mathbb{N}}[t - 1 + 2^{-n/2}, t)$. This verifies the induction step, so we are done by induction.

When f is not bounded, convergence trivially cannot be uniform: simple functions are bounded and hence the difference between an unbounded function and any simple function is unbounded.

In some texts you see authors implicitly assume that *any* approximation by simple func-

tions satisfies the final σ-algebra property. This is false; for example $\mathbf{1}_{[0,1]}$ is approximated by $\frac{n-1}{n}\mathbf{1}_{[0,0.5]} \cup \mathbf{1}_{(0.5,1]}$, but the former generates a smaller σ-algebra than the latter. The point is here this is a 'silly' approximation: there was no reason to split at 0.5; it is true that any 'sensible' approximation with sets defined by some inequality referring to f (as with our example) will satisfy the σ-algebra containment. Nevertheless, be somewhat careful with the literature here!

Theorem 12. *Let f be any $(X, \mathcal{F}_X, \mu)$-valued random variable on $(\Omega, \mathcal{F}, \mathbb{P})$, and let g be a $\mathbb{R}^{\pm\infty}$-valued random variable. Then the following are equivalent.*

- *There exists a $(\mathcal{F}_X, \mathcal{B}(\mathbb{R}^{\pm\infty}))$-measurable function $h : X \to \mathbb{R}^{\pm\infty}$ such that $g(\omega) = h\big(f(\omega)\big)$ for all $\omega \in \Omega$.*
- *g is $\sigma(f)$-measurable.*

Proof. One direction is easy. Suppose $g = h(f)$ for a measurable h. If $S \in \mathcal{B}(\mathbb{R}^{\pm\infty})$, then by definition $h^{-1}(S) \in \mathcal{F}_X$, by definition $f^{-1}\big(h^{-1}(S)\big) \in \sigma(f)$, and this verifies that g is $\sigma(f)$-measurable.

For the other direction, we need to do some work. The idea is to prove the result first for simple functions g, then for non-negative g, and finally in general.

Exercise 5. *Suppose g is a simple function. Prove h exists. Hint: Use Exercise 3; argue that you can assume that for any c there is at most one indicator function with coefficient c in the representation of g.*

Solution to Exercise 5:

By Exercise 3, we can assume $\sum_{i=1}^{t} \alpha_i \mathbf{1}_{A_i}$ is a disjoint indicator representation of g. Suppose that t is minimal subject to having a disjoint indicator representation. If for some $i \neq j$ we have $\alpha_i = \alpha_j$, we could replace $\alpha_i \mathbf{1}_{A_i} + \alpha_j \mathbf{1}_{A_j}$ with $\alpha_i \mathbf{1}_{A_i \cup A_j}$ is still a disjoint indicator representation, contradicting minimality of t; so the α_i are pairwise distinct.

Since the range of g is the numbers $R = \{0, \alpha_1, \ldots, \alpha_t\}$, and since g is $\sigma(f)$-measurable, the sets $A_r = g^{-1}(\{r\})$ for $r \in R$ partition Ω and are $\sigma(f)$-measurable. This says that for each $r \in R$ there is $B_r \in \mathcal{F}_X$ such that $A_r = f^{-1}(B_r)$. Since the A_r are pairwise disjoint, the sets $B_r \cap f(\Omega)$ are also pairwise disjoint (though the B_r might not be!). Since there are only finitely many sets B_r, we can obtain pairwise disjoint sets $B'_1 = B_1, B'_2 = B_2 \setminus B_1$ and so on which are still in $\mathcal{F}_X$ and which still satisfy $A_r = f^{-1}(B'_r)$ for all $r \in R$.

Define $h : X \to \mathbb{R}^{\pm\infty}$ by $h(x) = r$ if $x \in B'_r$ for some $r \in R$, and $h(x) = 0$ otherwise. By construction, this is the required Borel-measurable function.

Now, using Exercise 4, let $g_1, \cdots : \Omega \to \mathbb{R}$ be a pointwise monotone increasing sequence of functions which converge to the non-negative function g. By Exercise 5 we have a sequence of measurable functions $h_1, \cdots : X \to \mathbb{R}^{\pm\infty}$ such that $g_n = h_n(f)$ for each $n \in \mathbb{N}$. For any x in the range of f, the sequence $h_1(x), h_2(x), \ldots$ is monotone increasing since $g_1, \ldots$ is monotone increasing.

Letting $h = \sup_{n\in\mathbb{N}} h_n$, which is measurable by Theorem 5, it follows that for any x in the range of f we have $h(x) = \lim_{n\to\infty} h_n(x)$. Now for any $\omega \in \Omega$ we have $h(f(\omega) = \lim_{n\to\infty} h_n(f(\omega)) = \lim_{n\to\infty} g_n(\omega) = g(\omega)$, verifying $g = h(f)$ as required.

Finally, if g is a general measurable function, we write $g = g^+ - g^-$ and apply the previous part to g^+ and g^- separately to obtain h^+ and h^-. The required function is then $h^+ - h^-$. $\square$

This theorem holds for general measure spaces replacing $(\Omega, \mathcal{F}, \mathbb{P})$ (and the proof rather clearly works in general) but the stated case is the one we will use.

Apart from the 'philosophical' justification this gives for σ-algebras conveying information, this theorem will be useful for proving measurability.

6 Lebesgue integral

In our aim to generalise 'random variable' from the simple discrete case of Chapter 1, the next step is to say what the expectation of a random variable should be. For this, we need the Lebesgue integral.

The Riemann integral is defined for (some) functions from $\mathbb{R}$ to $\mathbb{R}$; it is supposed to be 'the area under the curve'. Its idea is based on the following two points:

(i) The area of an a by b rectangle is ab.

(ii) The integral should be linear: $\int(\alpha f + \beta g) = \alpha \int f + \beta \int g$.

(iii) If $f \leq g$ pointwise, then $\int f \leq \int g$ should be true.

We briefly recall (a version of) the Riemann integral: given a function $f : \mathbb{R} \to \mathbb{R}$, we find, for each $n \in \mathbb{N}$, approximations f_n^l and f_n^u to f which satisfy $f_n^l \leq f \leq f_n^u$ pointwise, and where f_n^l and f_n^u are both 'step functions'; that is, they are constant on each interval $[k/n, (k+1)/n)$ for $k \in \mathbb{Z}$. The obvious 'best' approximation we can use is to make f_n^l equal the infimum of $f(x)$ on $[k/n, (k+1)/n)$ and f_n^u the supremum (assuming these exist; if not the function fails to be Riemann integrable).

We can define $\int f_n^l$ and $\int f_n^u$: we just sum up the areas of all the rectangles as in (i) (these are called lower and upper Darboux sums, or sometimes Riemann sums). Suppose both these sums exist for all n, and their limits (these are series!) converge to a real number I. Then we define $\int f = I$, as the only option consistent with (iii).

This definition is somewhat simplified, and there are some difficulties with it. In particular, dealing with cancellation (if f has positive and negative values) is painful (we would like to have some monotonicity to make analyis theorems easier!) and dealing with series in the definition is similarly painful. The theory works more easily if we restrict our integral (for now) to only non-negative functions f, and to integrating over some bounded interval Ω in $\mathbb{R}$. Then the integrals of our approximations are finite sums of non-negative numbers; the cost is dealing with 'improper integrals (over $\mathbb{R}$, for example)' now needs taking limits explicitly, and if we want to integrate a general function we need to split $f = f^+ - f^-$ into the positive and negative parts.

There are still a few deficiencies of the Riemann integral. It does not deal with unbounded functions: though this can be 'fixed' (for example by looking at the limit of the integrals $\int \max(f, n)$ as $n \to \infty$; this is roughly what we will do for the Lebesgue integral shortly).

It does not deal with some functions we 'should' be able to integrate: for example $\int \mathbf{1}_{\mathbb{Q}\cap[0,1)}$ 'should' be 0, but the lower and upper sums do not converge to a common limit. Most importantly for us, it only applies to functions whose domain is (a subset of) $\mathbb{R}$.

The Lebesgue integral is based on replacing (i) with the more general

(iv) The integral $\int \mathbf{1}_A$ is $\mu(A)$.

Here we are implicitly working in a probability space (or a measure space with finite measure) $(\Omega, \mathcal{F}, \mathbb{P})$, and A should be $\mathcal{F}$-measurable for this to make sense. If we are working on $[0,1)$ with its standard Borel (or Lebesgue) measure, this clearly generalises (i): the area under $\mathbf{1}_{[t,t+a)}$ is an a by 1 rectangle.

Definition 13 (Lebesgue integral, non-negative functions)**.** *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be any probability space. If $g = \sum_{i=1}^{t} \alpha_i \mathbf{1}_{A_i}$ is any simple function, we define $\int_\Omega g \,\mathrm{d}\mathbb{P} = \sum_{i=1}^{t} \alpha_i \mathbb{P}(A_i)$.*

Let $f : \Omega \to \mathbb{R}^\infty_{\geq 0}$ be any non-negative measurable function. Then we define

$$\int_\Omega f \,\mathrm{d}\mathbb{P} = \sup_g \int_\Omega g \,\mathrm{d}\mathbb{P}\,,$$

where the supremum ranges over non-negative simple functions g such that $g \leq f$ pointwise.

Note that we *do* allow our integrand here to take value ∞. This is mainly so that we can later take limits of functions which are unbounded without worrying that we might leave the class of functions we can integrate; if a function takes value ∞ on a set of positive measure, its integral is ∞, so 'interesting' functions will only take value ∞ on a set of measure 0.

We normally won't want to stress the variable over which we're integrating: if we do, we'll write $\int_\Omega f(\omega) \,\mathrm{d}\mathbb{P}(\omega)$, or $\int_X f(x) \,\mathrm{d}x$, much as we would for Riemann integrals. We will usually want to be clear about the measure we're using to integrate, hence $\mathrm{d}\mathbb{P}$; if that is missing (as it was in the informal discussion!) then it is supposed to be clear from context. Finally, we may sometimes write $\int_A f \,\mathrm{d}\mathbb{P}$, where $A \in \mathcal{F}$ is some event: that is defined to mean $\int_\Omega f(\omega) \cdot \mathbf{1}_A(\omega) \,\mathrm{d}\mathbb{P}(\omega)$.

For us, the 'win' is that this notion naturally works for any probability space as the domain. It happens that we can now also integrate $\mathbf{1}_{\mathbb{Q}\cap[0,1)}$ over $[0,1)$, and the answer is 0, but realistically we probably never really wanted to integrate this function anyway. What is true (though not obvious) and useful is that the Lebesgue integral has some analytic properties the Riemann integral doesn't have, even if the domain is $[0,1)$. In particular it behaves more nicely if we want to integrate a sequence of functions and look at the limit function.

Exercise 6. *Prove that the Lebesgue integral is well-defined: i.e. that the integral of a simple function does not depend on its representation as a sum of indicator functions, and that the integral of a simple function as a sum matches the definition as a supremum. Hint: you may want to use Exercise 3.*

Prove that the Lebesgue integral is linear (i.e. $\int_\Omega (\alpha f + \beta g) \,\mathrm{d}\mathbb{P} = \alpha \int_\Omega f \,\mathrm{d}\mathbb{P} + \beta + \int_\Omega g \,\mathrm{d}\mathbb{P}$ whenever all three functions are non-negative). (Hint: First prove this for simple functions.)

Prove that if $0 \leq f \leq g$ pointwise, then $\int_\Omega f \,\mathrm{d}\mathbb{P} \leq \int_\Omega g \,\mathrm{d}\mathbb{P}$.

Solution to Exercise 6:

Let f be a simple function, given by an indicator representation $\sum_{i=1}^{t} \alpha_i \mathbf{1}_{A_i}$. Consider the construction of Exercise 3 which gives a disjoint interval representation

$$\sum_{I \subseteq [t]} \big(\sum_{j \in I} \alpha_j \big) \mathbf{1}_{A_I} \, .$$

We claim that this still gives the same numerical value for the integral. To see this, observe that the terms of the latter sum containing α_i are precisely

$$\alpha_i \sum i \in I \subseteq [t] \mathbf{1}_{A_I} = \alpha_i \mathbf{1}_B$$

where $B = \bigcup_{i \in I \subseteq [t]} A_I = A_i$ as required.

Now converting this representation to the block representation as described in the solution to Exercise 3 clearly does not change the integral (we just split some terms in the sum) and converting this to the slab representation again does not change the integral (we just sum up some terms). But the slab representation is $\sum_{i=1}^{r} v_i \mathbf{1}_{f^{-1}([v_i,\infty])}$, where the v_i are the different values the simple function takes: this by definition does not depend on the original indicator representation. (In some sense this whole thing is 'obvious'; it's just rearranging a finite sum.)

Next, suppose f is a simple function with $\sum_{i=1}^{t} \alpha_i \mathbf{1}_{A_i}$ a disjoint indicator representation. We have $\sup_{g \leq f} \int_\Omega g \, \mathrm{d}\mathbb{P} \geq \int_\Omega f \, \mathrm{d}\mathbb{P}$ (where both integrals should be understood as the definition for simple functions as a finite sum) because f is a simple function and is therefore in the supremum. And in fact we have equality: for suppose that $g \leq f$ is a simple function. Then $f - g$ is a non-negative simple function, whose integral is by definition non-negative: but we can write $f = g + (f - g)$ is an indicator representation of f. It follows

$$\int_\omega f \, \mathrm{d}\mathbb{P} \geq \int_\Omega g \, \mathrm{d}\mathbb{P} + \int_\Omega (f - g) \, \mathrm{d}\mathbb{P} \geq \int_\Omega g \, \mathrm{d}\mathbb{P}$$

again interpreting all integrals as the finite sum definition. This shows that $\int_\Omega f \, \mathrm{d}\mathbb{P}$ is an upper bound on the supremum, so we have equality. This verifies that the finite sum definition and the supremum definition coincide for simple functions, and hence the integral is well defined for all measurable functions.

Linearity for simple functions is trivial: taking the linear combination of indicator representations, by definition we obtain the linear combination of integrals. Similarly $\int_\Omega \alpha f \, \mathrm{d}\mathbb{P} = \alpha \int_\Omega f \, \mathrm{d}\mathbb{P}$ is trivial. Since $\alpha = 0$ is obvious, we can assume $\alpha > 0$. Now any simple function g in the former supremum corresponds to $\alpha^{-1} g$ appearing in the latter supremum, and vice versa, so that the latter supremum is α^{-1} times the former, and we obtain the required equality.

However what is left—to show for general measurable f, g that $\int_\Omega (f + g) \, \mathrm{d}\mathbb{P} = \int_\Omega f \, \mathrm{d}\mathbb{P} + \int_\Omega g \, \mathrm{d}\mathbb{P}$—is not nearly so easy. It is obvious that the RHS is a lower bound for the LHS: if we have any two simple functions in the RHS suprema, their sum is a simple function in the LHS supremum. But the other direction is false. The easiest route to a proof here is to look ahead to the Monotone Convergence Theorem, and check that we do not use this part

of linearity to prove it. (We do, in fact, use the scalar multiplication part, and linearity for simple functions!)

Let $f_1, \dots$ be a monotone increasing sequence of non-negative simple functions converging pointwise to f, and similarly $g_1, \dots$ monotone increasing simple functions converging pointwise to to g, using Exercise 4. Then $(f_n + g_n)$ is a monotone increasing sequence of simple functions converging to $f + g$. By Theorem 14, we have

$$\int_\Omega (f+g)\,\mathrm{d}\mathbb{P} = \lim_{n\to\infty} \int_\Omega (f_n + g_n)\,\mathrm{d}\mathbb{P} = \lim_{n\to\infty} \int_\Omega f_n\,\mathrm{d}\mathbb{P} + \lim_{n\to\infty} \int_\Omega g_n\,\mathrm{d}\mathbb{P} = \int_\Omega f\,\mathrm{d}\mathbb{P} + \int_\Omega g\,\mathrm{d}\mathbb{P}$$

as required.

Monotonicity of the integral is trivial: if $0 \le f \le g$, for any simple h such that $0 \le h \le f$ we also have $0 \le h \le g$ and hence the supremum in the definition of $\int_\Omega g\,\mathrm{d}\mathbb{P}$ is taken over a superset of the supremum in the definition of $\int_\Omega f\,\mathrm{d}\mathbb{P}$, so in particular the former supremum is at least as large as the latter.

Exercise 7. *Prove that for non-negative measurable functions f the following is true: $\int_\Omega f\,\mathrm{d}\mathbb{P} = 0$ if and only if $f = 0$ almost surely.*

Solution to Exercise 7:

Consider $A_n = \{\omega \in \Omega : f(\omega) \ge \frac{1}{n}\}$. If for some n we have $\mathbb{P}(A_n) > 0$, then by definition '$f = 0$ almost surely' is false, and $\int_\Omega f\,\mathrm{d}\mathbb{P} \ge \frac{1}{n}\mathbb{P}(A_n) > 0$ since $\frac{1}{n}A_n$ is a simple function which is bounded above by f.

If on the other hand we have $\mathbb{P}(A_n) = 0$ for all $n \in \mathbb{N}$, then by countable additivity,

$$0 = \mathbb{P}\Big(\bigcup_{n\in\mathbb{N}} A_n\Big) = \mathbb{P}\big(\{\omega \in \Omega : \exists n \in \mathbb{N}\,,\, f(\omega) \ge \tfrac{1}{n}\}\big) = \mathbb{P}\big(\{f \ne 0\}\big)\,,$$

and this says $f = 0$ almost surely. Furthermore, if $0 \le g \le f$ is any disjoint indicator representation and $\alpha \mathbf{1}_B$ is any term in the representation with $\alpha > 0$, then choosing $n > \alpha^{-1}$ we have $B \subseteq A_n$, so $\mathbb{P}(B) = 0$. It follows that $\int_\Omega g\,\mathrm{d}\mathbb{P} = 0$, so $\int_\Omega f\,\mathrm{d}\mathbb{P}$ is the supremum of $\{0\}$ which is 0.

One thing that appears to be 'missing' is the guarantee we have with Riemann integration that we get the 'right' integral: we do not have a matching upper bound as with the upper Darboux sums. But this is easy to recover using Exercise 4.

Exercise 8. *Suppose $f : \Omega \to \mathbb{R}$ is a bounded non-negative measurable function. Use Exercise 4 to construct a sequence of simple functions $g_1, \dots$ such that $0 \le f \le g_n$ holds pointwise for each n, and such that $\lim_{n\to\infty} \int_\Omega g_n\,\mathrm{d}\mathbb{P} = \int_\Omega f\,\mathrm{d}\mathbb{P}$.*

(Hint: consider a sequence of simple functions which are bounded above by f pointwise and whose integrals converge to $\int_\Omega f\,\mathrm{d}\mathbb{P}$. Use the fact that the maximum of two simple functions is a simple function.)

Prove that the approximating functions $f_1, \dots$ of f constructed by Exercise 4 satisfy $\lim_{n\to\infty} \int_\Omega f_n\,\mathrm{d}\mathbb{P} = \int_\Omega f\,\mathrm{d}\mathbb{P}$.

Prove that the last statement still holds if $f : \Omega \to \mathbb{R}^\infty_{\ge 0}$ is any non-negative measurable function (not necessarily bounded).

Solution to Exercise 8:

Let f be a bounded non-negative measurable function; suppose B is a real number such that $f \leq B$. Let $f_1, \ldots$ be the approximating monotone increasing sequence of simple functions guaranteed by Exercise 4, and for each $\varepsilon > 0$ let $N(\varepsilon)$ be minimal be such that if $n > N(\varepsilon)$ we have $|f - f_n| < \varepsilon$ pointwise (the uniform convergence guarantee).

By definition of the supremum, there is a sequence $h_1, \ldots$ of simple functions bounded above by f such that $\lim_{n\to\infty} \int_\Omega h_n \, \mathrm{d}\mathbb{P} = \int_\Omega f \, \mathrm{d}\mathbb{P}$. Let for each n the function $f'_n = \max(f_n, h_1, \ldots, h_n)$. This is a simple function bounded above by f, which satisfies $|f - f'_n| < \varepsilon$ whenever $n > N$, and the sequence also satisfies $\lim_{n\to\infty} \int_\Omega f'_n \, \mathrm{d}\mathbb{P} = \int_\Omega f \, \mathrm{d}\mathbb{P}$ by monotonicity of the integral.

Let $g_n = f_n + \varepsilon_n \mathbf{1}_\Omega$ where ε_n is taken minimal such that $N(\varepsilon_n) < n$; if this does not exist then $\varepsilon_n = B$. By construction, $\varepsilon_n \to 0$ as $n \to \infty$, and by construction $f \leq g_n$ holds for all $n \in \mathbb{N}$. But also by construction,

$$\lim_{n\to\infty} \int_\Omega g_n \, \mathrm{d}\mathbb{P} = \lim_{n\to\infty} \int_\Omega f_n \, \mathrm{d}\mathbb{P} + \lim_{n\to\infty} \varepsilon_n = \int_\Omega f \, \mathrm{d}\mathbb{P}$$

as required.

For each $n \in \mathbb{N}$, define $g'_n = f_n + \varepsilon_n \mathbf{1}_\Omega$. Then we have pointwise $f \leq g'_n \leq g_n$, so by monotonicity of the integral and the Sandwich Theorem we have $\lim_{n\to\infty} \int_\Omega g'_n \, \mathrm{d}\mathbb{P} = \int_\Omega f \, \mathrm{d}\mathbb{P}$. But since $\int_\Omega f_n \, \mathrm{d}\mathbb{P} = \int_\Omega g'_n \, \mathrm{d}\mathbb{P} - \varepsilon_n$, we obtain $\lim_{n\to\infty} \int_\Omega f_n \, \mathrm{d}\mathbb{P} = \int_\Omega f \, \mathrm{d}\mathbb{P}$ as required.

For unbounded f, the easiest route to showing that the approximating sequence of Exercise 4 has the required convergence of the integral is to show that it holds for any monotone increasing approximating sequence: this is the Monotone Convergence Theorem. Again, this requires checking that Exercise 8 is not used in the proof!

Something we should always have in mind as a 'danger' is the standard example $n^2 \mathbf{1}_{[1/n, 1/(n-1)]}$ of functions which converge pointwise to the constant 0 function but whose integrals do not converge to 0. Because of this example, we can't instantly say that any sequence of functions which approximates f pointwise will also have integral converging to $\int_\Omega f \, \mathrm{d}\mathbb{P}$, and indeed it won't be true if the approximating functions are not restricted (we can just add the above bad example to f..!). The following theorem shows that this danger does not occur if we use monotone increasing approximations bounded above by f.

Exercise 9. *Let $F_1 \subseteq F_2 \subseteq \ldots$ be a sequence of events in $(\Omega, \mathcal{F}, \mathbb{P})$ whose union is Ω. Prove that for any $A \in \mathcal{F}$ we have*

$$\lim_{n\to\infty} \int_\Omega \mathbf{1}_{F_n} \mathbf{1}_A \, \mathrm{d}\mathbb{P} = \mathbb{P}(A) \, .$$

Deduce that for any simple f we have

$$\lim_{n\to\infty} \int_\Omega \mathbf{1}_{F_n} f \, \mathrm{d}\mathbb{P} = \int_\Omega f \, \mathrm{d}\mathbb{P} \, .$$

Solution to Exercise 9:

We can rephrase the first statement as: prove that $\lim_{n\to\infty} \mathbb{P}(F_n \cap A) = \mathbb{P}(A)$. Since $F_1 \cap A, F_2 \cap A, \ldots$ is a nested sequence of sets whose union is A, this follows immediately from continuity of measure.

The required statement for simple f follows immediately from the first part together with linearity of the integral for simple functions. (Note: it's important we only need this for simple functions! otherwise our proof of general linearity would be circular.)

Theorem 14 (Monotone Convergence Theorem)**.** *Let $f_1, \ldots$ be a sequence of measurable functions on $(\Omega, \mathcal{F}, \mathbb{P})$ such that $0 \le f_1 \le f_2 \le \ldots$ holds pointwise. Then $\lim_{n\to\infty} f_n$ is a measurable function, and we have*

$$\lim_{n\to\infty} \int_\Omega f_n \, \mathrm{d}\mathbb{P} = \int_\Omega f \, \mathrm{d}\mathbb{P} \,.$$

Proof. That $f := \lim_{n\to\infty} f_n$ exists is a consequence of the fact that the functions are pointwise monotone increasing: for any given ω, either $\big(f_n(\omega)\big)_{n\in\mathbb{N}}$ is bounded, in which case as a monotone increasing bounded sequence it tends to a (real number) limit, or it is not bounded in which case by definition the limit is ∞. That it is measurable is immediate from Theorem 5.

Similarly, $\lim_{n\to\infty} \int_\Omega f_n \, \mathrm{d}\mathbb{P}$ exists as the terms of the sequence are, by Exercise 6, monotone increasing.

Exercise 10. *Prove that $\int_\Omega \lim_{n\to\infty} f_n \, \mathrm{d}\mathbb{P}$ exists and is an upper bound for $\big(\int_\Omega f_n \, \mathrm{d}\mathbb{P}\big)_{n\in\mathbb{N}}$.*

Solution to Exercise 10:

The existence of the integral $\int_\Omega f \, \mathrm{d}\mathbb{P}$ is definitional: every non-negative measurable function has an integral.

That the integral is an upper bound for the given set of extended real numbers is easy: since $f_1, \ldots$ is a monotone increasing sequence, for any given n we have $f_n \le f$ pointwise, so by monotonicity of the integral we have $\int_\Omega f_n \, \mathrm{d}\mathbb{P} \le \int_\Omega f \, \mathrm{d}\mathbb{P}$ as required.

To complete the proof, we need to argue that $\int_\Omega f_n \, \mathrm{d}\mathbb{P}$ becomes arbitrarily close to $\int_\Omega f \, \mathrm{d}\mathbb{P}$. The basic idea here is that as n grows, the measure of points ω such that $f_n(\omega)$ is much smaller than $f(\omega)$ must tend to zero, and these points can only be responsible for a tiny difference in the integral. To formalise this, though, we go by way of a simple function approximation to f.

Let g be a simple function which takes values in $\mathbb{R}_{\ge 0}$ with $g \le f$ pointwise. Fix $0 < \gamma < 1$. For each $n \in \mathbb{N}$, let $A_n = \{\omega \in \Omega \,:\, f_n(\omega) \ge \gamma g(\omega)\}$.

Because the f_n are monotone increasing, $A_1 \subseteq A_2 \subseteq \ldots$ are nested. We claim their union is Ω: if not, then there is some ω which is in none of these sets, so $f_n(\omega) < \gamma g(\omega)$ holds for all $n \in \mathbb{N}$, so $\lim_{n\to\infty} f_n(\omega) \le \gamma g(\omega) < g(\omega) \le f(\omega)$, contradicting that f is the pointwise limit.

Observe that by definition of A_n and monotonicity of the integral, for each n we have

$$\int_\Omega f_n \, \mathrm{d}\mathbb{P} \ge \int_\Omega f_n \mathbf{1}_{A_n} \, \mathrm{d}\mathbb{P} \ge \int_\Omega \gamma g \mathbf{1}_{A_n} \, \mathrm{d}\mathbb{P} = \gamma \int_\Omega g \mathbf{1}_{A_n} \, \mathrm{d}\mathbb{P} \,,$$

where the equality is by linearity of the integral. Since g is a simple function, by Exercise 9 the limit of the RHS as $n \to \infty$ is $\gamma \int_\Omega g \, \mathrm{d}\mathbb{P}$, and it follows that

$$\lim_{n\to\infty} \int_\Omega f_n \, \mathrm{d}\mathbb{P} \geq \gamma \int_\Omega g \, \mathrm{d}\mathbb{P} \, .$$

Since this bound holds for all $\gamma < 1$, we obtain $\lim_{n\to\infty} \int_\Omega f_n \, \mathrm{d}\mathbb{P} \geq \int_\Omega g \, \mathrm{d}\mathbb{P}$.

Letting $g_1, \ldots$ be a sequence of simple functions taking values in $\mathbb{R}_{\geq 0}$ with $g_m \leq f$ pointwise and with $\lim_{m\to\infty} \int_\Omega g_m \, \mathrm{d}\mathbb{P} = \int_\Omega f \, \mathrm{d}\mathbb{P}$ (which exists by Exercise 8), we obtain $\lim_{n\to\infty} \int_\Omega f_n \, \mathrm{d}\mathbb{P} \geq \int_\Omega g_m \, \mathrm{d}\mathbb{P}$ for all m, and in particular

$$\lim_{n\to\infty} \int_\Omega f_n \, \mathrm{d}\mathbb{P} \geq \int_\Omega f \, \mathrm{d}\mathbb{P} \, . \qquad \square$$

The following is an easy corollary of the Monotone Convergence Theorem.

Theorem 15 (Fatou's Lemma)**.** *Let $f_1, \ldots$ be any non-negative measurable functions. Then we have*

$$\int_\Omega \liminf_{n\to\infty} f_n \, \mathrm{d}\mathbb{P} \leq \liminf_{n\to\infty} \int_\Omega f_n \, \mathrm{d}\mathbb{P} \, .$$

Exercise 11. *Prove this: recall that $\liminf_{n\to\infty} f_n$ means $\lim_{n\to\infty} \inf_{k\geq n} f_k$, and for any functions f_k this limit is a limit of pointwise monotone increasing functions. You will need to use monotonicity of the integral in addition to the Monotone Convergence Theorem.*

Solution to Exercise 11:

We have

$$\int_\Omega \liminf_{n\to\infty} f_n \, \mathrm{d}\mathbb{P} = \lim_{n\to\infty} \int_\Omega \inf_{k\geq n} f_k \, \mathrm{d}\mathbb{P} \leq \lim_{n\to\infty} \inf_{n\geq k} \int_\Omega f_k \, \mathrm{d}\mathbb{P} = \liminf_{n\to\infty} \int_\Omega f_n \, \mathrm{d}\mathbb{P}$$

where the first equality is the Monotone Convergence Theorem (which applies for the reason given in the exercise statement), the inequality is monotonicity of the integral, and the final inequality is the definition of limit infimum.

A corollary of Theorem 15, sometimes called the Reverse Fatou Lemma, is that if $f_1, \ldots$ are pointwise bounded above by some measurable g with $\int_\Omega g \, \mathrm{d}\mathbb{P} < \infty$, then we have

$$\int_\Omega \limsup_{n\to\infty} f_n \, \mathrm{d}\mathbb{P} \geq \limsup_{n\to\infty} \int_\Omega f_n \, \mathrm{d}\mathbb{P} \, .$$

This follows by applying Fatou's Lemma to the functions $g - f_1, g - f_2, \ldots$. We have to be slightly careful here: if $g(\omega) = f_n(\omega) = \infty$ for some ω and n, we should take $(g - f_n)(\omega) = 0$. Since the set of ω such that $g(\omega) = \infty$ is necessarily of measure zero, this change does not affect the integrals. We use the fact that g has finite integral to justify rearranging the resulting

$$\int_\Omega \liminf_{n\to\infty} (g - f_n) \, \mathrm{d}\mathbb{P} \leq \liminf_{n\to\infty} \int_\Omega (g - f_n) \, \mathrm{d}\mathbb{P} \, .$$

We complete this chapter by giving the Lebesgue integral in general, and a corresponding statement allowing us to interchange pointwise limit and the integral.

Definition 16 (Lebesgue integral)**.** *Given a probability space* $(\Omega, \mathcal{F}, \mathbb{P})$ *and a measurable function* $f : \Omega \to \mathbb{R}^{\pm\infty}$*, let* $f^+ = \max(f, 0)$ *and* $f^- = -\max(-f, 0)$ *be respectively the positive and negative parts of* f*.*

Suppose that at most one of $\int_\Omega f^+ \,\mathrm{d}\mathbb{P}$ *and* $\int_\Omega f^- \,\mathrm{d}\mathbb{P}$ *is* ∞ *(if both are, then the integral is undefined). Then we define* $\int_\Omega f \,\mathrm{d}\mathbb{P} = \int_\Omega f^+ \,\mathrm{d}\mathbb{P} - \int_\Omega f^- \,\mathrm{d}\mathbb{P}$*.*

It is straightforward to check that this integral is also linear and monotonic, and that it agrees with Definition 13 on the integral of simple functions.

If you want only to integrate functions from $\mathbb{R}$ to $\mathbb{R}$, the Riemann integral handles an infinite domain of integration better: there are functions which are not Lebesgue integrable (because both the positive and negative parts integrate to ∞) but which have an improper Riemann integral. Still, you usually will be taught to take limits of a finite range, and this you can do with the Lebesgue integral too. The Lebesgue integral handles unbounded functions correctly (but the Riemann integral can be made to do this too, again by taking limits). And most of the functions you really want to integrate are in fact both Riemann and Lebesgue integrable.

However, for proofs, even in this setting the Lebesgue integral has a big advantage: you can under very mild conditions swap integral and limit—the point being that the pointwise limit of measurable functions is still measurable, whereas the pointwise limit of Riemann integrable functions is not always Riemann integrable. If you're working with Riemann integrals, you need to write a separate proof that in your particular case the pointwise limit is integrable, or to do more analysis to avoid working with a 'limit integral' that doesn't exist. Since swapping integration and limits is something we want to do a lot, it is extremely helpful to be able to do it without needing much extra work.

If f is a measurable function on $[a, b] \subseteq \mathbb{R}$ which is Riemann integrable, then it is almost immediate that the Riemann and Lebesgue integrals agree. The step functions corresponding to a Darboux lower and upper sum are simple functions whose integrals are the lower and upper sums; by monotonicity of the Lebesgue integral, they are lower and upper bounds on the integral, and since the sums converge to a common limit as the partition becomes finer, this common limit (which by definition is the Riemann integral) must also be the Lebesgue integral. To complete the proof that the Lebesgue integral really generalises the Riemann integrable, we need the following.

Exercise 12. *Prove that if* $f : [a, b] \to \mathbb{R}$ *is Riemann integrable with finite integral, then* f *is measurable (with the Lebesgue measure* λ *on* $[a, b]$*).*

Solution to Exercise 12:

For each $k \geq 1$, let P_k be a partition of $[a, b]$ into intervals; suppose P_{k+1} refines P_k (i.e. each set of P_{k+1} is contained in one set of P_k). Let $L(P_k, f)$ be the step function taking value $\inf_{x \in I} f(x)$ on every $I \in P_k$, and $U(P_k, f)$ the step function taking value $\sup_{x \in I} f(x)$ on each $I \in P_k$. Suppose that this sequence of partitions witnesses the Riemann integral: that is, the limits as $k \to \infty$ of $\int_{[a,b]} L(P_k, f) \,\mathrm{d}x$ and $\int_{[a,b]} U(P_k, f) \,\mathrm{d}x$ exist and are equal. Note that here it does not matter which integral we are considering: the Riemann and Lebesgue integrals agree for step functions (as is easy to check). Furthermore (we can assume, removing if necessary the first few k) the functions $L(P_k, f)$ and $U(P_k, f)$ take values in $\mathbb{R}$ because the Riemann integral is finite.

Let $L(f)$ be the pointwise limit of $L(P_k, f)$ as $k \to \infty$; this exists because the sequence is (by definition) pointwise increasing and bounded above by f. Similarly define $U(f)$ as the pointwise limit of the $U(P_k, f)$; this is a monotone decreasing sequence bounded below by f, so $L(f) \leq f \leq U(f)$ pointwise.

The functions $L(f)$ and $U(f)$ are measurable functions, as limits of step functions (which are simple so clearly measurable), so they are Lebesgue integrable. By monotonicity of the integral, we have

$$\int_{[a,b]} L(P_k, f)\,\mathrm{d}x \leq \int_{[a,b]} L(f)\,\mathrm{d}x \leq \int_{[a,b]} U(f)\,\mathrm{d}x \leq \int_{[a,b]} U(P_k, f)\,\mathrm{d}x\,,$$

and as $k \to \infty$ the outer pair of integrals are assumed (Riemann integrability) to converge to the same number: so the middle inequality is an equality, and we have $\int_{[a,b]} U(f) - L(f)\,\mathrm{d}x = 0$. Since $U(f) \leq f \leq L(f)$, so $U(f) - L(f)$ is a non-negative function, and Exercise 7 now tells us $U(f) = L(f)$ almost surely; in other words, there is a set $A \subseteq [a,b]$ with $\lambda(A) = 0$ such that outside A we have $L(f) = f = U(f)$.

Let F be any Borel-measurable set in $\mathbb{R}$. Because $U(f)$ is measurable, we have $(U(f))^{-1}(F) \in \mathcal{L}([a,b])$. Because f and $U(f)$ agree outside of A, we have

$$(U(f))^{-1}(F) \setminus A = f^{-1}(F) \setminus A$$

and hence $f^{-1}(F) = \big((U(f))^{-1}(F) \setminus A\big) \cup A'$ for some $A' \subseteq A$. Because $\lambda(A) = 0$, and because we are working with the Lebesgue measure, which is complete, the set A' is measurable with $\lambda(A') = 0$, and hence $f^{-1}(F)$ is measurable, as required.

It's reasonable to wonder whether we really need to use Lebesgue measure in this theorem: it's clear where we use it, which is to take $A' \subseteq A$ and get a measurable set. The answer is we do: if $f = \mathbf{1}_{C'}$ where C' is any subset of the Cantor set C, then f is Riemann integrable (with integral zero). This is relatively easy to prove: it's enough to show $\mathbf{1}_C$ has this property. But we 'know' that there are subsets of C which are not Borel-measurable; these give us indicator functions which are not measurable if we use the Borel measure on $[a,b]$. So we indeed need to go to Lebesgue measure.

Often definitions of Riemann integral don't allow unbounded functions anyway, but if we do allow such functions, it's straightforward to see that for this exercise we need the condition that the Riemann integral is finite: otherwise we can take any function which is integrable and with infinite integral on $(0,1]$ (for example, $x \to x^{-2}$) and combine it with $\mathbf{1}_B$ for any $B \subseteq [-1,0]$ at all to obtain a function on $[-1,1]$ which is Riemann integrable. But if B is not measurable, neither is the function. The point where we needed finite Riemann integral in the proof is where we subtracted the inner integral from the outer to get 0; this isn't valid if both sides are ∞ (or $-\infty$).

Historically, Lebesgue defined, for a non-negative measurable function f, the integral $\int_\Omega f\,\mathrm{d}\mathbb{P}$ to be the (usual improper Riemann) integral

$$\int_{t=0}^{\infty} \mathbb{P}\big(\{x \,:\, f(x) \geq t\}\big)\,\mathrm{d}t\,.$$

When you see comments about 'partitioning the range not the domain', what's being referred to is the (improper) Riemann sum you get from this definition.

Exercise 13. *Prove that this definition gives the same integral as Definition 13.*

Solution to Exercise 13:

It's straightforward to see the two definitions agree on simple functions: given a simple function, consider its slab representation (as discussed in the solution to Exercise 3). Lebesgue's definition of the integral of this function is the Riemann integral of a step function, which is just the sum of the product of step height and width—which is exactly the same as our definition of Lebesgue integral for such a function.

It's easy to see Lebesgue's definition satisfies monotonicity of the integral: this follows from the same property for the Riemann integral and monotonicity of measures under subsets. It's also easy to see that all measurable functions are Lebesgue integrable in both senses: we need it to guarantee the measure in Lebesgue's definition is always defined, but then the Riemann integral is the integral of a monotone decreasing function, which is known (and this is quite easy to prove) to exist. This is already enough to prove that Lebesgue's definition gives an integral of any measurable $f \geq 0$ which is at least our definition of $\int_\Omega f \, \mathrm{d}\mathbb{P}$, and (by Exercise 8) that they coincide if f is bounded. The only thing left is to deal with unbounded f. For each $N \in \mathbb{N}$, let $f_N = \min(f, N)$ which is a bounded function: then using the definition of the improper Riemann integral we have

$$\begin{aligned}
\int_{t=0}^{\infty} \mathbb{P}\big(\{x \,:\, f(x) \geq t\} \, \mathrm{d}t &= \lim_{N\to\infty} \int_{t=0}^{N} \mathbb{P}\big(\{x \,:\, f(x) \geq t\}\big) \, \mathrm{d}t \\
&= \lim_{N\to\infty} \int_{t=0}^{N} \mathbb{P}\big(\{x \,:\, f_N(x) \geq t\}\big) \, \mathrm{d}t \\
&= \lim_{N\to\infty} \int_{t=0}^{\infty} \mathbb{P}\big(\{x \,:\, f_N(x) \geq t\}\big) \, \mathrm{d}t \\
&= \lim_{N\to\infty} \int_{\Omega} f_N \, \mathrm{d}\mathbb{P} \\
&= \int_{\Omega} f \, \mathrm{d}\mathbb{P} \,,
\end{aligned}$$

where the penultimate equality is because both definitions of the Lebesgue integral agree for the bounded function f_N, and the final line is the Monotone Convergence Theorem.

In practice, though, Definition 13 is much easier to work with.

Finally, the Dominated Convergence Theorem gives a general condition allowing us to swap limit and integral.

Theorem 17 (Dominated Convergence Theorem)**.** *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space; let $f_1, \dots$ be a sequence of measurable functions from Ω to $\mathbb{R}^{\pm\infty}$ which converges pointwise to f. Suppose that there exists some $g : \Omega \to \mathbb{R}^{\infty}_{\geq 0}$ such that $|f_1|, |f_2|, \dots \leq g$ pointwise, and such that $\int_\Omega g \, \mathrm{d}\mathbb{P} < \infty$. Then we have*

$$\int_{\Omega} f \, \mathrm{d}\mathbb{P} = \lim_{n\to\infty} \int_{\Omega} f_n \, \mathrm{d}\mathbb{P} \quad \textit{and} \quad \lim_{n\to\infty} \int_{\Omega} |f - f_n| \, \mathrm{d}\mathbb{P} = 0 \,.$$

Proof. By the Triangle Inequality we have $|f - f_n| \leq |f| + |f_n| \leq 2g$ pointwise, and since $f_1, \dots$ tends pointwise to f, we have pointwise $\limsup_{n\to\infty} |f - f_n| = 0$.

We therefore have

$$0 \leq \Big| \int_\Omega f \,\mathrm{d}\mathbb{P} - \int_\Omega f_n \,\mathrm{d}\mathbb{P} \Big| = \Big| \int_\Omega (f - f_n) \,\mathrm{d}\mathbb{P} \Big| \leq \int_\Omega |f - f_n| \,\mathrm{d}\mathbb{P} \tag{1}$$

where the equality is by linearity of the integral, and the inequality by monotonicity (twice, once applied to the integral and once to its negative).

Taking the limit supremum, by the Reverse Fatou Lemma (the corollary of Theorem 15), which applies since $|f - f_n| < 2g$ pointwise, we have

$$\limsup_{n\to\infty} \int_\Omega |f - f_n| \,\mathrm{d}\mathbb{P} \leq \int_\Omega \limsup_{n\to\infty} |f - f_n| \,\mathrm{d}\mathbb{P} = \int_\Omega 0 \,\mathrm{d}\mathbb{P} = 0\,.$$

Since the integrals on the LHS are bounded below by 0 (their integrand is non-negative) we have $\lim_{n\to\infty} \int_\Omega |f - f_n| \,\mathrm{d}\mathbb{P} = 0$, which by (1) completes the proof. □

The conclusion that the integral of $|f - f_n|$ tends to zero is (usefully) stronger than the fact that we can interchange the integral and limit. The difference is that (in principle) we might find the integrals happen to be equal because although the integrands are often far apart, the difference swaps signs and the positive and negative parts cancel. In such a situation, we would have no reason to expect that $\int_\Omega hf \,\mathrm{d}\mathbb{P}$ is close to $\int_\Omega hf_n \,\mathrm{d}\mathbb{P}$, where h is some measurable function: it might spoil the cancellation. In fact, if h is unbounded even under the assumptions of Theorem 17 we have no reason to expect to be able to swap integral and limit.

However if h is a bounded measurable function, say $-C \leq h \leq C$ pointwise, the fact $\lim_{n\to\infty} \int_\Omega |f - f_n| \,\mathrm{d}\mathbb{P} = 0$ is enough to bring h in:

$$0 \leq \Big| \int_\Omega hf \,\mathrm{d}\mathbb{P} - \int_\Omega hf_n \,\mathrm{d}\mathbb{P} \Big| \leq \int_\Omega |hf - hf_n| \,\mathrm{d}\mathbb{P} = \int_\Omega |h||f - f_n| \,\mathrm{d}\mathbb{P} \leq \int_\Omega C|f - f_n| \,\mathrm{d}\mathbb{P}$$

and the right hand integral by assumption tends to 0 as $n \to \infty$, so $\int hf \,\mathrm{d}\mathbb{P} = \lim_{n\to\infty} hf_n \,\mathrm{d}\mathbb{P}$.

This is a (rather cheap!) example of the following general idea: good control over 'moments' of a single random variable can help us understand things about more complicated objects, like products of two random variables. We'll see more of this in what follows.

7 Expectation, moments and independence

Definition 18 (Expectation, moments, centred moments, variance)**.** *Let* $(\Omega, \mathcal{F}, \mathbb{P})$ *be a probability space, and let* $X : \Omega \to \mathbb{R}^{\pm\infty}$ *be a random variable. Then, if it exists, we say the* expectation of X *is* $\mathbb{E}(X) = \int_\Omega X \,\mathrm{d}\mathbb{P}$*. This is often called the* mean of X*.*

Given any $s \in \mathbb{R}$*, if it exists, the* sth moment of X *is* $\mathbb{E}(X^s)$*. So the first moment of* X *is its expectation; usually (not always!) we are interested in the case* $s \in \mathbb{N}$*.*

The sth centred moment of X*, again if it exists, is* $\mathbb{E}\big((X - \mathbb{E}(X))^s\big)$*, and in particular the second centred moment* $\mathbb{E}\big((X - \mathbb{E}(X))^2\big)$ *is usually called the* variance of X*, written* $\mathrm{Var}(X)$*. Centred moments are considered undefined if* $\mathbb{E}(X)$ *is not a real number.*

Theorem 19 (Markov's inequality)**.** *Let X be any non-negative random variable, and suppose $\mathbb{E}(X) < \infty$ exists. Then for any $t > 0$ we have*

$$\mathbb{P}(X \geq t) \leq \tfrac{\mathbb{E}(X)}{t}\,.$$

Exercise 14. *Prove this. Hint: consider $A = \{\omega : X(\omega) \geq t\}$.*

Solution to Exercise 14:

Let $A = \{\omega : X(\omega) \geq t\}$ which is measurable (it's the inverse image of $[t, \infty]$).

$$\mathbb{E}(X) = \int_\Omega X \,\mathrm{d}\mathbb{P} \geq \int_\Omega t \cdot \mathbf{1}_A \,\mathrm{d}\mathbb{P} = t\mathbb{P}(A)\,,$$

where the inequality is monotonicity of the integral and the second equality is the definition of the integral for simple functions. Rearranging, we obtain Markov's inequality. Here we used the condition $X \geq 0$ to guarantee $X \geq t \cdot \mathbf{1}_A$ outside of A.

Theorem 20 (Chebyshev's inequality)**.** *Let X be any non-negative random variable, and suppose $\mathrm{Var}(X) < \infty$ exists. Then for any $t > 0$ we have*

$$\mathbb{P}\big(|X - \mathbb{E}(X)| \geq t\big) \leq \tfrac{\mathrm{Var}(X)}{t^2}\,.$$

Proof. Since $\mathbb{E}(X)$ exists, $(X - \mathbb{E}(X))^2$ is a random variable by Theorem 5, we assumed its expectation $\mathrm{Var}(X)$ exists, and it is obviously non-negative. Applying Markov's inequality, we have

$$\mathbb{P}\big((X - \mathbb{E}(X))^2 \geq t^2\big) \leq \tfrac{\mathrm{Var}(X)}{t^2}$$

and since $(X - \mathbb{E}(X))^2 \geq t^2$ holds if and only if $|X - \mathbb{E}(X)| \geq t$, the theorem follows. $\square$

We should read these two inequalities as telling us: control of the expectation lets us show a non-negative random variable is unlikely to get very large. Control of the variance lets us show a random variable is unlikely to get far from its expectation.

It's not hard to come up with 'higher moments' (i.e. applying to nth moments or centred moments for larger integers n) versions of these statements: broadly, control over higher moments usually translates to stronger bounds on 'unlikely', though one should be careful about the fact that odd moments can be small due to cancellation! In this course, we'll avoid needing control of higher moments.

Definition 21 (Independence: of σ-algebras, random variables)**.** *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. If $\mathcal{G}_1, \ldots, \mathcal{G}_n$ are any finite collection of σ-algebras on Ω contained in $\mathcal{F}$, we say they are* independent *if the following holds. For any $G_1 \in \mathcal{G}_1, \ldots, G_n \in \mathcal{G}_n$, the events $G_1, \ldots, G_n$ are independent.*

We say random variables $X_1, \ldots, X_n$ are independent if $\sigma(X_1), \ldots, \sigma(X_n)$ are independent.

An infinite collection of σ-algebras, or random variables, are independent if every finite sub-collection is independent.

It's important to remember that even though the probability measure $\mathbb{P}$ only appears once in the above definition, it is nevertheless critical: independence depends on $\mathbb{P}$. Two events can be independent with respect to $\mathbb{P}$ but not independent with respect to another probability measure $\mathbb{Q}$ on $\mathcal{F}$, even if those measures are 'nicely' related: in fact, 'typically' changing measures will change which pairs of events are independent, and so also which random variables are independent.

Theorem 22 (Properties of expectation)**.** *Let X and Y be two random variables on $(\Omega, \mathcal{F}, \mathbb{P})$, and suppose $\mathbb{E}(|X|), \mathbb{E}(|Y|) < \infty$. For any $\alpha, \beta \in \mathbb{R}$ we have linearity of expectation*

$$\mathbb{E}\big(\alpha X + \beta Y\big) = \alpha \mathbb{E}(X) + \beta \mathbb{E}(Y)\,.$$

If in addition X and Y are independent, we have $\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y)$.

Proof. The first part is a direct consequence of linearity of the Lebesgue integral: we use $\mathbb{E}(|X|), \mathbb{E}(|Y|) < \infty$ to ensure that in the definition of the integral we do not attempt to write or cancel $\infty - \infty$.

Exercise 15. *Prove multiplicativity of expectation for independent random variables. (Hint: first consider the case that X and Y are non-negative: use Exercise 4 and the Monotone Convergence Theorem. Then use this case, together with the Dominated Convergence Theorem, to deal with the general case.)*

Solution to Exercise 15:

Suppose first X and Y are non-negative independent random variables, and we have $\mathbb{E}(X), \mathbb{E}(Y) < \infty$.

Let $f_1, \dots$ be a monotone increasing sequence of $\sigma(X)$-measurable simple functions tending to X, and $g_1, \dots$ similarly for Y, as given by Exercise 4.

We claim $f_n g_n$ is a sequence of monotone increasing simple functions which tend to XY. That the functions are simple is 'obvious': let $f_n = \sum_{\alpha_i} \mathbf{1}_{A_i}$ and $g_n = \sum_{\beta_j} \mathbf{1}_{B_j}$, and we have

$$f_n g_n = \sum_{i,j} \alpha_i \beta_j \mathbf{1}_{A_i \cap B_j}$$

by expanding the brackets (which we can do because this is a finite sum). Similarly, that we have the required pointwise convergence is a standard fact of analysis.

Because $\sigma(X)$ and $\sigma(Y)$ are independent, and f_n and g_n are measurable with respect to these two σ-algebras respectively, we have $\mathbb{P}(A_i)\mathbb{P}(B_j) = \mathbb{P}(A_i \cap B_j)$ for all the terms, and hence

$$\int_\Omega f_n g_n \,\mathrm{d}\mathbb{P} = \int_\Omega f_n \,\mathrm{d}\mathbb{P} \cdot \int_\Omega g_n \,\mathrm{d}\mathbb{P}\,.$$

Taking the limit as $n \to \infty$, and applying the Monotone Convergence Theorem to swap all three limits with the integrals, we have

$$\int_\Omega XY \,\mathrm{d}\mathbb{P} = \int_\Omega X \,\mathrm{d}\mathbb{P} \cdot \int_\Omega Y \,\mathrm{d}\mathbb{P}\,.$$

Having proved this, we can now apply almost identical logic with general random variables: Exercise 4 gives us a sequence of simple functions which tends to (but is not monotone

increasing!) X by applying it to X^+ and X^- separately, and similarly for Y: what we get replacing 'monotone increasing' is pointwise convergence with $-|X| \leq f_n \leq |X|$ holding pointwise, and similarly for Y. The equality of integrals for the simple approximations holds, and we have pointwise $|f_n| \leq |X|$, $|g_n| \leq |Y|$ and $|f_n g_n| \leq |X||Y|$. Since we assumed $\mathbb{E}(|X|), \mathbb{E}(|Y|) < \infty$ and by the non-negative case proved $\mathbb{E}(|X||Y|) < \infty$, we can apply the Dominated Convergence Theorem in place of the Monotone Convergence Theorem to justify swapping limits and integrals in the final step.

□

It's finally convenient to note a fact about expectation and convex functions. We say $f : I \to \mathbb{R}$, where I is any interval in $\mathbb{R}$, is *convex* if $f(\alpha x + (1-\alpha)y) \leq \alpha f(x) + (1-\alpha)f(y)$ holds for all $x, y \in I$ and $\alpha \in [0,1]$. In other words, f is convex if whenever we draw a chord between two points on the function's graph, f lies below the chord.

Theorem 23 (Jensen's inequality; reverse Jensen)**.** *Let X be a real-valued random variable on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$, and let f be a convex function whose domain contains the range of X. Then we have*

$$\mathbb{E}\big(f(X)\big) \geq f\big(\mathbb{E}(X)\big)\,.$$

If the range of X is contained in $[a,b]$ where $a, b \in \mathbb{R}$, and α is such that $\mathbb{E}(X) = \alpha a + (1-\alpha)b$, then we have

$$\mathbb{E}\big(f(X)\big) \leq \alpha f(a) + (1-\alpha)f(b)\,.$$

The first inequality is Jensen's inequality; the second is sometimes called the reverse Jensen inequality.

Proof. A (somewhat) standard piece of real analysis shows the following: for any x_0 in the interior of I (i.e. there is some open interval containing x_0 which is in I), there exists $c \in \mathbb{R}$ such that for all $x \in I$ we have $c(x - x_0) + f(x_0) \leq f(x)$; such a c is called a *subgradient at* x_0.

One way to prove this is: look at the set F_1 of c such that the inequality fails for some $x < x_0$. If $f(x) \leq f(x_0)$ then $1 \in F_1$; if not, $2\frac{f(x)-f(x_0)}{x-x_0} \in F_1$, so F_1 is non-empty. Similarly F_2 the set of c such that the inequality fails for some $x > x_0$ is non-empty. It is straightforward to check (by convexity) that all elements of F_1 are strictly less than any element of F_2, and both sets are open: so any c with $\sup(F_1) \leq c \leq \inf(F_2)$ has the desired property.

Now let $x_0 = \mathbb{E}(X)$. If x_0 is not in the interior of I, then by Exercise 7, $X = x_0$ holds almost surely and the inequality is trivial. So suppose x_0 is in the interior of I, and let c be a subgradient at x_0.

By monotonicity and then linearity of the integral, we have

$$\int_\Omega f(X)\,\mathrm{d}\mathbb{P} \geq \int_\Omega c(X - x_0) + f(x_0)\,\mathrm{d}\mathbb{P} = c\int_\Omega X\,\mathrm{d}\mathbb{P} - cx_0 + f(x_0) = cx_0 - cx_0 + f(x_0) = f\big(\mathbb{E}(X)\big)$$

which is Jensen's inequality as required.

The reverse Jensen inequality is a bit simpler: by convexity, we have $f(a) + \frac{x-a}{b-a}\big(f(b) - f(a)\big) \geq f(x)$ for all $x \in [a,b]$. So

$$\int_\Omega f(X)\,\mathrm{d}\mathbb{P} \leq \int_\Omega f(a) + \tfrac{X-a}{b-a}\big(f(b) - f(a)\big)\,\mathrm{d}\mathbb{P} = f(a) + \big(f(b) - f(a)\big)\tfrac{\mathbb{E}(X)-a}{b-a}$$

and substituting $\mathbb{E}(X) = \alpha a + (1-\alpha)b$ we get the desired inequality. □

The following is officially a theorem, though it's often presented as being the definition (it's not!).

Theorem 24 (Law of the Unconscious Statistician)**.** *Let X be a real-valued random variable on $(\Omega, \mathcal{F}, \mathbb{P})$, and let $F(x) = \mathbb{P}([-\infty, x))$.*

Suppose there exists a non-negative $f : \mathbb{R} \to \mathbb{R}$ such that $F(x) = \int_{-\infty}^x f(y)\,\mathrm{d}y$.

Then for any measurable $g : \mathbb{R} \to \mathbb{R}$, provided the integrals exist we have

$$\mathbb{E}\big(g(X)\big) = \int_{-\infty}^{\infty} g(x)f(x)\,\mathrm{d}x\,,$$

where these integrals should be interpreted as the usual improper integral, i.e. given by taking limits as $N \to \infty$ of $\int_{-N}^x f(y)\,\mathrm{d}y$ and $\int_{-N}^N g(x)f(x)\,\mathrm{d}x$ respectively.

This proof isn't examinable (but you might be interested).

Proof. We prove the statement for non-negative g, and this implies the general result by the usual trick of writing $g = g^+ - g^-$ (here our conditions ensure the integrals are finite).

The aim is to approximate g by monotone increasing simple functions, argue that we have equality for simple functions, and use the Monotone Convergence Theorem to take limits.

Let $g_1, \ldots$ be any monotone increasing sequence of non-negative simple functions converging pointwise to g; let $g_n = \sum_i \alpha_{i,n}\mathbf{1}_{A_{i,n}}$ where the $A_{i,n}$ are a finite collection of pairwise disjoint Borel set in $\mathbb{R}$ for each fixed n. Without loss of generality, we may assume that each g_n is supported on $[-n, n]$ (otherwise we would replace g_n by $g_n\mathbf{1}_{[-n,n]}$), so $A_{i,n} \subseteq [-n, n]$.

Then

$$\mathbb{E}\big(g(X)\big) = \lim_{n\to\infty} \mathbb{E}\big(g_n(X)\big) = \lim_{n\to\infty} \sum_i \alpha_{i,n}\mathbb{P}\big(X^{-1}(A_{i,n})\big)\,,$$

where the first equality is the MCT and the second equality is using the definition of the integral for simple functions.

We now claim $\mathbb{P}\big(X^{-1}(A_{i,n})\big) = \int_{x=-\infty}^{\infty} f(x)\mathbf{1}_{A_{i,n}}\,\mathrm{d}x$ for any bounded Borel set $A_{i,n}$. Sup-

posing the claim is true for now, we have

$$\begin{aligned}
\mathbb{E}\big(g(X)\big) = \lim_{n\to\infty} \sum_i \alpha_{i,n} \mathbb{P}\big(X^{-1}(A_{i,n})\big) &= \lim_{n\to\infty} \sum_i \alpha_{i,n} \int_{x=-\infty}^{\infty} f(x) \mathbf{1}_{A_{i,n}} \,\mathrm{d}x \\
&= \lim_{n\to\infty} \int_{x=-n}^{n} f(x) g_n(x) \,\mathrm{d}x \\
&= \lim_{n\to\infty} \lim_{m\to\infty} \int_{x=-m}^{m} f(x) g_n(x) \,\mathrm{d}x \\
&= \lim_{m\to\infty} \lim_{n\to\infty} \int_{x=-m}^{m} f(x) g_n(x) \,\mathrm{d}x \\
&= \lim_{m\to\infty} \int_{x=-m}^{m} f(x) g(x) \,\mathrm{d}x \\
&= \int_{x=-\infty}^{\infty} f(x) g(x) \,\mathrm{d}x \,,
\end{aligned}$$

as we wanted. Here several steps need justification. On the first line we just substituted in our integral formula, and the second uses (finite!) linearity of the integral, together with the fact that g_n is supported on $[-n, n]$. The third line is a triviality: if $m \geq n$ the inner limit becomes a constant sequence. The fourth line is justified because we can swap limits for a doubly monotone sequence: for any $b_{m,n}$ satisfying $b_{n,m} \leq b_{n+1,m}, b_{n,m+1}$ for all n, m we have $\lim_{n\to\infty} \lim_{m\to\infty} b_{n,m} = \lim_{m\to\infty} \lim_{n\to\infty} b_{n,m}$. This is an (easy) fact of real analysis, but note it's false without the monotonicity condition!

The fifth line is the Monotone Convergence Theorem: we are applying it to the finite measure space $[-m, m]$ with Borel measure, which is by scaling equivalent to the probability measure case we proved. (We didn't prove it for the σ-finite case of $\mathbb{R}$!)

Finally, the sixth line is the definition of the improper integral.

It remains to prove our Claim: for any bounded $A \in \mathcal{B}(\mathbb{R})$, we have $\mathbb{P}\big(X^{-1}(A)\big) = \int_{-\infty}^{\infty} f(x) \mathbf{1}_A \,\mathrm{d}x$.

This is true by definition if $A = (-\infty, x)$ for any $x \in \mathbb{R}$, and by finite linearity of the integral it is true for any finite union of half-open intervals $[a, b)$.

Suppose now A is a countable union of disjoint half-open intervals $A_1, \ldots$ contained in $[-m, m)$ for some $m \in \mathbb{N}$. We have

$$\mathbb{P}\big(X^{-1}(A)\big) = \lim_{n\to\infty} \mathbb{P}\big(X^{-1}(A_1 \cup \cdots \cup A_n)\big) = \lim_{n\to\infty} \int_{x=-n}^{n} f(x) \mathbf{1}_{A_1 \cup \cdots \cup A_n} \,\mathrm{d}x = \int_{x=-n}^{n} f(x) \mathbf{1}_A \,\mathrm{d}x$$

where the first equality is continuity of measure, the second is finite linearity of the integral, and the third is the Monotone Convergence Theorem.

Finally, let A be any Borel set contained in $[-m, m)$ for some $m \in \mathbb{N}$. The finite case of the Carathéodory Extension Theorem, applied (as in Notes 1) to construct a unique measure from the pre-measure on the algebra generated by half-open intervals from the distribution function F, states that $\mathbb{P}(X^{-1}(A))$ is equal to the outer pre-measure of A; in other words, there is a sequence $B_1, \ldots,$ each of which is a countable union of disjoint half-open intervals, each of which contains A, and such that $\mathbb{P}(X^{-1}(A)) = \lim_{n\to\infty} \mu(B_n)$. Without loss of

generality (replace B_n by $B_n \cap [-m, m)$) we may assume that all the sets B_n are contained in $[-m, m]$. Then we have

$$\mathbb{P}(X^{-1}(A)) = \lim_{n\to\infty} \mathbb{P}(X^{-1}(B_n)) = \lim_{n\to\infty} \int_{x=-m}^{m} f(x)\mathbf{1}_{B_n}\,\mathrm{d}x \geq \int_{x=-m}^{m} f(x)\mathbf{1}_A$$

where the inequality is by monotonicity of the integral. Now the same inequality applies to $A' = [-m, m) \setminus A$, so we have

$$\begin{aligned}\int_{x=-m}^{m} f(x)\,\mathrm{d}x &= \mathbb{P}\big(X^{-1}([-m, m))\big) = \mathbb{P}(X^{-1}(A)) + \mathbb{P}(X^{-1}(A')) \\ &\leq \int_{x=-m}^{m} f(x)\mathbf{1}_A\,\mathrm{d}x + \int_{x=-m}^{m} f(x)\mathbf{1}_{A'}\,\mathrm{d}x = \int_{x=-m}^{m} f(x)\,\mathrm{d}x\end{aligned}$$

where the final equality is linearity of the integral. So the inequality must be an equality, in particular

$$\mathbb{P}(X^{-1}(A)) = \int_{x=-m}^{m} f(x)\mathbf{1}_A\,\mathrm{d}x = \int_{x=-\infty}^{\infty} f(x)\mathbf{1}_A\,\mathrm{d}x$$

as claimed. □

We could've shortened this proof somewhat by proving the Monotone Convergence Theorem for σ-finite measure spaces (instead of just finite ones); that follows by basically the same limit-swapping argument we used above. Overall, the proof is rather 'straightforward'; we do the usual thing of approximating our general measurable function by simple functions, and simple functions are easy to work with, though we need a little care to explain why the measure of any Borel set is given by the corresponding integral. (This is often skipped over in textbooks, though it does really need a proof! An alternative is to show that the integral defines a measure on Borel sets and agrees with the measure $\mathbb{P}(X^{-1}(\cdot))$ on a generating set of the Borel σ-algebra, then appeal to the uniqueness part of Carathéodory. This is shorter if one has the MCT for σ-finite measure spaces; if not, we need to repeat the limit-swapping argument and it ends up longer.)

Something you should notice, if you compare this proof to standard 'undergraduate' proofs that rely on the Riemann integral, is that we haven't had to do anything too difficult, but somehow our result works for general g, as opposed to requiring (for instance) g^{-1} to exist and be monotone. What's really going on here is that to work with any integral, we need to know something about $A = g^{-1}([p, q])$ for intervals $[p, q]$. We relied on this being a Borel set, because we can deal with them; this is why our condition is just that g is measurable. If you want to work with the Riemann integral, having a Borel set is not good enough: A could be a very nasty set, perhaps with $\mathbf{1}_A$ not being Riemann integrable even though g is. Conditions like saying g^{-1} exists and is monotone are there to avoid this happening. In short, this is a proof where we really benefit a lot from working with the strong notion of Lebesgue integration.

If you want to, you can generalise this further by replacing the integral with the probability density function g by a Riemann-Stieltjes integral $\mathrm{d}F$; but we won't need that in this course.